AUDITOR RESIGNATIONS VERSUS DISMISSALS: AN EXAMINATION OF THE DIFFERENTIAL EFFECTS ON MARKET LIQUIDITY AND TRADING ACTIVITY

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ABSTRACT

This study addresses whether an auditor change (a resignation or a dismissal) mitigates information asymmetry as measured by market liquidity or trading activity. For auditor dismissals our results show no effect on our sample firms' market liquidity or trading activity. By contrast, for auditor resignation firms, the market liquidity tests indicate that the 12 month period preceding the 8K filing is characterized by rising information asymmetry. Also, our trading activity analysis suggests that the 8K auditor resignation filing is informative for individual investors but fails to support such informativeness for institutional investors. Our findings lend support for the SEC's decision to differentiate between auditor resignations and dismissals in the 8K. However, the resignation announcement does not appear to decrease information asymmetry subsequent to the 8K filing, which is inconsistent with the presumed SEC objective of maintaining public confidence in the securities markets as a level playing field by mitigating information asymmetry.

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INTRODUCTION

This paper is motivated by the question, does the additional disclosure about whether an auditor change is a resignation or a dismissal mitigate information asymmetry? Over the years, the Securities and Exchange Commission (SEC) has sought to discourage "opinion shopping" by requiring firms to publicly disclose an auditor change in their Form 8K filings. However, it was only in 1988 that disclosure of whether the auditor resigned or was dismissed was made mandatory.

Although prior research indicates a differential stock price response to auditor resignation versus dismissal announcements, thus suggesting the usefulness of distinguishing between the two in the 8K filings, usefulness is a necessary but not a sufficient condition for mandatory disclosure (Lev, 1988). Rather, SEC disclosure regulation is motivated primarily by its mandate to protect the investing public and maintain public confidence in the securities markets as a level playing field by mitigating information asymmetry (Lev, 1988; Levitt, 1998; Shefrin & Statman, 1992).

Information asymmetry implies an information gap between some (informed) investors and others based on private information search, i.e. the ability of informed investors to obtain information that is not publicly available or to extract private information from publicly available data. Differential access to information allows informed investors to increase their wealth at the expense of uninformed investors and is perceived by the U.S. Congress and the SEC to be unfair (Beaver, 1998). As a practical matter, unequal access to information can have adverse social consequences in the form of reduced liquidity (higher transaction costs and thinner markets) as market makers respond defensively to a perceived increase in information asymmetry (Lev, 1988).

An auditor resignation, unlike a dismissal, is likely to be auditor (rather than client) initiated. Business Week (1993) and MacDonald (1997) indicate that resignations are related to unfavorable events within the client firm and may be triggered by one or more of the following: the auditor suspects management fraud (or otherwise distrusts client management), the client has weak internal controls, or the client's financial health is deteriorating. Potentially, a resignation is indicative of future investor losses that might be realized as and when the full extent of the client's problems (such as fraudulent financial statements or financial distress) are eventually revealed.

Consistent with the notion that lawsuits against auditors are almost always triggered by investor losses (Lys & Watts, 1994), DeFond et al. (1997) and Krishnan and Krishnan (1997) suggest that auditor resignations are motivated by the desire to avoid costly litigation and may be triggered by negative private information

(obtained during the course of an audit) indicating a deterioration in the client's risk characteristics. Since a resignation indicates an increased probability that the client has a hidden risk, Bockus and Gigler (1998) predict analytically a decline in the client's stock price following an auditor resignation.

Krishnan and Krishnan (1997) indicate that auditor resignation firms are more susceptible to events that trigger litigation than firms which dismiss their auditors. Their findings suggest that sophisticated investors may be able to use litigation prediction models to anticipate auditor resignations. Potentially, the time period leading up to the 8K auditor resignation filing is likely to be characterized by increasing information asymmetry between informed and uninformed investors. As noted previously, the increase in information asymmetry can be expected to have an observable effect in terms of reduced market liquidity.

Further, as pointed out by Hakansson (1981), it takes resources to acquire private information. Potentially, institutional investors are likely to possess the necessary resources to acquire private pre-disclosure information and are therefore less likely to be dependent on SEC mandated disclosures. By contrast, individual investors are much more likely to be dependent on SEC mandated disclosures. Hence, relative to their institutional counterparts, individual investors (because of their greater dependence on SEC mandated disclosure) can be expected to display heightened trading activity at the time of the 8K auditor resignation filing.

In this study, we utilize intraday stock transactions data to examine whether the auditor resignation/dismissal announcement has differential effects on market liquidity (as measured by the effective spread, the adverse selection component of the spread, and quoted depth) over a time frame beginning 12 months before and ending 2 months after the announcement. In addition, we utilize the methodology described in Cready and Mynatt (1991) to examine the trading response surrounding the time of the 8K filing to identify the type of investor responding to the resignation/dismissal announcement. Our sample consists of all auditor change firms during 1995 and 1996.

Consistent with the notion that auditor resignations may be anticipated by informed investors (Krishnan & Krishnan, 1997), our results suggest decreasing market liquidity (increasing information asymmetry) for the resignation firms during the 12 months preceding the 8K filing. Although the public announcement of the auditor resignation appears to halt the trend towards lower liquidity, it does *not* improve liquidity. Also, the trading-based analysis suggests a strong response surrounding the disclosure of an auditor resignation but little or no response surrounding a dismissal. Our analysis indicates that this trading response occurs largely within lower size transactions, suggesting that

individual investors rely on the 8K resignation announcement while institutional investors rely more on private pre-disclosure information.

In the next section, we review prior research on auditor changes. Subsequent sections discuss our sample and data, the research methodology and design, and the empirical findings. The final section contains our concluding remarks.

PRIOR RESEARCH

At this time, there is no general theory to explain auditor changes. Among the reasons cited in the literature for an auditor switch are an auditor-client mismatch, changes in management, a need for non-audit advisory services that the prior auditor was unable to provide, lower fees, and a search for a more prestigious (Big Six) auditor as a means of reducing agency costs (Burton & Roberts, 1967; Bedingfield & Loeb, 1974; Eichenseher & Shields, 1983; Johnson & Lys, 1990; McConnell, 1984). These reasons suggest that auditor changes may have legitimate motives and may not be intended to subvert the integrity of the independent audit. Of course, any of the cited reasons may simply mask a situation representing a disagreement between the client and the auditor. Indeed, client-auditor disagreements may go unreported in the 8K since such disclosure may not be in the best interest of either party (DeFond & Jiambalvo, 1993).

One might expect the growing number and complexity of accounting and financial reporting regulations over the past decade to increase the potential for technical disputes between client management and the auditor. Given the freedom to choose, the client retains the prerogative to dismiss the current auditor and hire a more compliant replacement auditor. From a policy perspective, the matter of concern to the SEC is that a high frequency of auditor changes can have a negative effect on *all* auditors to be less critical of their clients' accounting decisions and thus damage the perceived reliability of *all* audited financial statements. Still, the SEC's current disclosure policies are intended to balance conflicting objectives by impeding opinion shopping but not discouraging justifiable auditor changes.[1]

As noted previously, the SEC's requirement for disclosing auditor resignations is relatively recent. To our knowledge, with two exceptions (DeFond et al., 1997; Wells & Loudder, 1997), prior market-based research (Eichenseher et al., 1990; Fried & Schiff, 1981; Johnson & Lys, 1990; Klock, 1994) has examined auditor changes in general rather than differentiate between auditor resignations and dismissals. These prior studies have attempted to establish the *usefulness* of the 8K disclosures by examining the effect on the stock price (using standard abnormal stock return methodology) to the disclosure in the 8K

of an auditor switch. A significant negative price effect in the days surrounding the filing of the 8K would suggest that investors perceive auditor switches to represent opportunistic behavior by management. Alternatively, no price reaction would suggest that opinion shopping is not perceived to be a pervasive problem (which could be because of the public's faith in the SEC's policing of auditor changes or despite the SEC's attention to the issue). In general, prior research suggests no significant price effect on average to auditor changes reported in the 8K.

Wells and Loudder (1997) examined the stock price effects surrounding auditor resignations (rather than auditor switches in general). They report a significant negative price reaction in the two days subsequent to the disclosure of an auditor resignation in the 8K.[2] In further cross-sectional analyses, the negative price effect was *not* associated with the size of the firm or any other information (other than the resignation itself) disclosed in the 8K. Wells and Loudder (1997) conclude that "it is the disclosure of the resignation itself, not any other related information, that is causing the results" (p. 143). Collectively, prior research suggests that although there is no price reaction associated with auditor changes in general, there is a negative price effect associated with auditor resignations. Overall, the implication is that auditor dismissals are viewed neutrally and not as a symptom of opinion shopping intended to undermine the integrity of the independent audit; by contrast, resignations appear to be viewed negatively by investors.

DeFond et al. (1997) suggest that auditor resignations are motivated by litigation risk. Krishnan and Krishnan (1997) indicate that resignation companies differ systematically from dismissal companies along financial distress and other risk characteristics that proxy the probability of litigation. Because a reported resignation could cause a stock price decline and trigger a lawsuit against both the client and the auditor, Krishnan and Krishnan (1997, fn. 15) suggest that both parties have an incentive to avoid reporting a resignation. Consequently, resignations may be underreported and some resignations may be misclassified as dismissals. Such misclassifications would in fact militate against finding significant differences between resignation and dismissal firms.

Still, Krishnan and Krishnan indicate that a model utilized in the prior literature (Stice, 1991) to predict litigation against audit firms may be useful in predicting auditor resignations. Similarly, Menon and Williams (1999) report results consistent with the notion that "clients who are dropped by their auditors tend to have more characteristics associated with auditor-related litigation than clients who dismiss their auditors" (p. 121). These findings suggest that informed investors may be able to anticipate auditor resignations based on pre-disclosure private information search. Hence, as discussed previously, the time period

preceding the resignation announcement is likely to be characterized by increasing information asymmetry. Also, relative to their institutional counterparts, individual investors are more likely to be dependent on SEC mandated disclosures and to display increased trading activity in response to the 8K auditor resignation filing.

As described in the next section, we differentiate between auditor resignation and dismissal announcements and evaluate them from the perspective of both market liquidity and trading activity across different investor groups. Our capital market analyses is therefore different from, yet complementary to, the prior research discussed in this section.

DATA AND METHODOLOGY

For this study, our sample consists of firms that had an auditor change in the two years 1995 and 1996. We included all firms which during 1995 or 1996 disclosed an auditor change in their 8K filings (as obtained from the LEXIS database) and for which we were able to obtain intraday trading data from the Trade and Quotations (TAQ) database published by the New York Stock Exchange (NYSE). The TAQ database provides intraday transactions data for both exchange-listed and NASDAQ firms.

We chose the years 1995 and 1996 for our study since: (1) our research design (as discussed below) required trading data for the 12 months preceding and 2 months following the auditor change, and (2) the database available to us contained intraday trading data for 1994 through early 1997 only. There were a total of 140 auditor change (22 auditor resignation and 118 auditor dismissal) firms in our sample. Admittedly, the sample of auditor resignation firms is relatively small. Unfortunately, due to resource and data constraints we were not able to expand the study beyond 1995 and 1996. Hence, it is important to recognize that our results must be interpreted with great caution.

As noted previously, in our study we differentiate between auditor resignations and dismissals. From a market liquidity perspective, we evaluate whether the 8K auditor resignation/dismissal filing has differential effects on information asymmetry by examining the behavior of three market liquidity measures (the effective half-spread, the adverse selection component of the spread, and quoted depth) for the sample firms for a period beginning 12 months before and ending 2 months after the release of the 8K. Consistent with the prior market microstructure literature (discussed below), we utilize these liquidity measures as observable proxies for information asymmetry. From a trading activity perspective, we evaluate the trading response of different investor groups to the resignation/dismissal announcement based on the methodology described in Cready and Mynatt (1991).

Differential Effects on Market Liquidity

Observable Measures of Information Asymmetry

According to the prior market microstructure literature (Bagehot, 1971; Callahan et al., 1997; Coller & Yohn, 1997; Copeland & Galai, 1983; Glosten & Milgrom, 1985), the role of the market maker (dealer) is to provide liquidity or immediacy by standing ready to trade at his/her declared bid and ask quotes. The sellers and buyers who trade with the market maker can be liquidity motivated (i.e. seek to convert securities into cash and vice versa) or information motivated (i.e. possess private information). The dealer is not thought to do any private information search (such as analyzing the financial statements of the companies in whose shares he/she trades) and is viewed as relatively uninformed (Kim & Verrecchia, 1994, p. 44). Hence, although the market maker gains from liquidity motivated traders (because of the bid-ask spread) he/she loses to information motivated traders (due to adverse selection). Thus, individuals with private information profit at the expense of the market maker who, in turn, profits from those who are liquidity motivated.

To survive in the business, the dealer's gains from transacting with the liquidity motivated must exceed the losses from transacting with the information motivated. Normally, the market maker cannot distinguish ex ante between the two types of transactors, and dealers' inventories typically decrease (increase) before an increase (decrease) in the equilibrium price suggesting a loss due to the lag in adjusting the bid-ask spread (Stoll, 1976).

Understandably, market makers undertake defensive measures to protect themselves against losses sustained in trading with those who possess private information. The market microstructure literature has sought to understand these protective actions in order to predict how information asymmetry affects financial markets (O'Hara, 1995). As discussed below, these protective measures include widening of the bid-ask spread and/or reducing the quoted depth, either of which ceteris paribus is consistent with decreased market liquidity.

The effective half-spread. One observable effect of information asymmetry is a wider bid-ask spread. In particular, market makers are thought to widen the spread between the bid and ask quotes for firms with greater information asymmetry. A wider quoted spread is deemed necessary to earn additional profits in transactions with liquidity traders in order to offset losses in transactions with informed traders. Seminal work which has focused on this effect includes Copeland and Galai (1983), Glosten and Milgrom (1985), and Easley and O'Hara (1987).

Prior accounting research (e.g. Greenstein and Sami 1994) has used the width of the *quoted* bid-ask spread as an observable proxy for the extent of

information asymmetry. However, the problem with the use of the *quoted* bid-ask spread as a measure of information asymmetry is that a large fraction of actual trades take place inside the quoted spread (Lee & Ready, 1991). Hence, the *effective* spread accounts for trades inside the posted quotes and is a more accurate reflection of a trader's execution costs than the *quoted* spread. Also, since the spread applies to a round trip transaction (two trades), Huang and Stoll (1996) utilize half-spreads as a measure of information asymmetry and the market maker's loss to informed investors per trade. Consistent with Huang and Stoll (1996), we define the *effective* half-spread per share as $|p_t - m_t|$, where p_t is the trade price at time t and m_t is the midpoint (average) of bid and ask quotes at time t. We use the effective half-spread deflated by the midpoint of the quoted bid and ask prices at the time of the trade as our first measure of information asymmetry. The *greater* the effective half-spread (variable EHS), the *lower* the market liquidity.

The adverse selection component of the spread. Huang and Stoll (1996) define the adverse selection component of the spread as the difference between the *effective* half spread and the realized half spread. As noted previously, the *effective* half spread takes into account the fact that a large fraction of actual trades take place inside the quoted spread. The *realized* spread measures the dealer's realized revenues on the basis of the post-trade price reversal and is defined as the average difference between the price at which a market maker sells at a given point in time and the price at which he/she buys at an earlier point in time. On average, a market maker does not realize the *effective* spread since prices typically move against him/her, i.e. reverse by falling after a dealer buy and rising after a dealer sale, due to adverse information possessed by informed traders. Hence, the realized spread is generally less than the effective spread and the excess of the *effective* half-spread over the realized half-spread provides an estimate of the market maker's average loss per share on a transaction with information motivated traders. To survive in the business, the market maker has to recover this loss (by widening the total spread) in transactions with liquidity motivated traders. Hence, the adverse selection (information) component of the spread represents the market maker's loss to informed traders (Huang & Stoll, 1996). For this reason, the adverse selection component of the spread is a more focused measure of information asymmetry than the *effective* half-spread discussed previously.

As noted previously, we define the *effective* half-spread per share as $|p_t - m_t|$, where p_t is the trade price at time t and m_t is the midpoint (average) of bid and ask quotes at time t. Consistent with Huang and Stoll (1996), we define the *realized* half-spread as $(p_{t+\tau} - p_t)$ when the transaction at time t occurs at the bid

quote and $-(p_{t+\tau} - p_t)$ when the transaction at time t occurs at the ask quote, where p_t is the transaction price at time t and $p_{t+\tau}$ is the transaction price of the first trade occurring at least 5 minutes after the initial trade at time t. Huang and Stoll (1996) choose 5 minutes as the period of time necessary for an offsetting trade to occur and demonstrate that their empirical results are insensitive to the use of alternative time periods. For each firm in our sample, we calculate the average adverse selection component of the spread by taking the excess of the effective half-spread over the realized half-spread per share on each stock transaction, scaling (deflating) the excess by the midpoint of the quoted bid and ask prices at the time of the trade, and averaging across all transactions for each month. The *greater* the adverse selection component of the spread (variable ASCS), the *lower* the market liquidity.

Quoted depth. Lee et al. (1993) suggest that a more complete measure of market liquidity requires an examination of not only bid-ask spreads but also depth, i.e. the number of shares the market maker is willing to trade at each quote. They suggest that faced with informed trading, the market maker can increase spread as well as reduce depth, i.e. both spread and depth can be adjusted to actively manage the risk of loss to informed traders. For example, reducing quoted depth (i.e. offering to trade a fewer number of shares at each quoted price) protects the market maker by reducing the maximum loss that he/she can sustain on a single trade. Lee et al. (1993) indicate that a combination of a wider spread and a lower depth is sufficient to infer decreased liquidity. Ceteris paribus, the lower the depth (variable DEPTH), the lower the market liquidity.

Models and Independent Variables

Trend in liquidity. In this analysis, we focus on the trend in liquidity in the 12 months leading up to the 8K auditor change filing:

$$\ln(y_{it}) = \alpha_{i0} + \alpha_1 \ln(NTRADE_{it}) + \alpha_2 \ln(PRICE_{it}) + \alpha_3 \ln(VAR_{it}) + \alpha_4 MONTH_{it} + \alpha_5(MONTH_{it} * RESIGN_{it}) + \varepsilon_{it} \quad (1)$$

Model 1 is a fixed-effects panel model and includes dummy variables for each firm in the sample. Hence, the intercept is free to vary across firms and measures the baseline level of liquidity in firm i in the 12th month preceding the auditor change (the subscripts i and t denote firm and month, respectively). In model 1, the firm-specific intercepts control for differences in the baseline level of information asymmetry (liquidity) across firms. Following prior research, we

use the log form of the variables (except the indicator variables) in the regressions. The dependent variable $\ln(y_{it})$ consists of each of the three liquidity measures discussed previously (EHS, ASCS, and DEPTH) and is regressed on a set of variables that control for the market maker's inventory holding, order processing, and potential adverse selection costs. Consistent with the prior market microstructure literature (e.g. Brennan & Subrahmanyam, 1995; Easley et al., 1996; Stoll, 1978; Venkatesh & Chiang, 1986), these control variables are the number of daily transactions (NTRADE), the price of the stock (PRICE), and the variance of daily stock returns (VAR).

For the regressions with the spread measures (EHS and ASCS) as the dependent variables, the predicted signs for NTRADE, PRICE, and VAR (following the prior literature) are negative, negative, and positive, respectively. First, the greater the number of transactions (NTRADE), the shorter the dealer's inventory holding period, the lower the opportunity cost of funds tied up in inventory, and the lower the spread. Also, the probability of informed trading is lower in higher volume stocks (Easley et al., 1996). Hence, the greater the number of transactions, the lower the probability of informed trading, and the lower the required spread to compensate the dealer for potential adverse selection costs. For these reasons, variable NTRADE is expected to have a negative sign.

Second, a decrease in the stock price implies lower holding costs (and a lower dollar spread) since the same number of shares may now be held in inventory with fewer dollars invested. However, since our spread measures are obtained after deflating the dollar spread by price, a higher stock price (PRICE) implies a lower spread. Hence, variable PRICE is expected to have a negative sign. Finally, the greater the variability in the stock price, the greater the risk that the price might fall, and the greater the spread that is necessary to compensate the market maker for potential inventory holding losses. Also, the greater the volatility in the stock price, the greater the market maker's anticipated losses to informed traders since the probability of obtaining larger deviations between the price based on public information and the price based on private information is greater for firms with more volatile stocks. Hence, the greater the return variance, the greater the market maker's potential adverse selection costs. For these reasons, variable VAR is expected to have a positive sign.

As noted previously, for a given level of market liquidity, depth is expected to vary inversely with spread. For this reason, in the model with DEPTH as the dependent variable, the predicted signs for the control variables (with the exception of PRICE) are expected to be the opposite of the predicted signs in the models with the spread measures (EHS or ASCS) as the dependent variables. Since DEPTH is not deflated by the price of the stock, the expected sign

for variable PRICE is negative since (as discussed previously) a lower stock price implies lower inventory holding costs. Hence, in the regression with DEPTH as the dependent variable, the predicted signs for NTRADE, PRICE, and VAR are positive, negative, and negative, respectively.

In model 1, MONTH is a categorical variable which takes a value of 0 in month −12 relative to the 8K auditor change filing month, 1 in month −11, and so on up to a value of 11 in the month preceding the 8K filing. Since the dependent variable is log-transformed, the slope coefficient 4 measures the monthly rate of change in the dependent variable for our sample firms. As discussed previously, auditor switches in general may be legitimately motivated and thus not associated with a change in information asymmetry; hence, we do not predict the sign of the MONTH variable.

Finally, MONTH*RESIGN is an interaction variable between variable MONTH and RESIGN (which is a dummy variable equal to 1 for an auditor resignation firm and 0 otherwise). The sum of the slope coefficients (α_4+α_5) measures the monthly rate of change in the dependent variable for resignation firms. As discussed previously, the ability of sophisticated investors to anticipate auditor resignations (based on publicly known litigation risk factors) implies also the ability to anticipate investor losses and to avoid those losses. Hence, we hypothesize that for resignation firms the time period before the 8K filing is likely to be characterized by rising information asymmetry and decreasing liquidity. In the regressions with the spread measures (EHS and ASCS) as the dependent variable, we hypothesize a positive sign for the MONTH*RESIGN interaction variable indicating a widening spread for resignation firms relative to dismissal firms. In the regression with DEPTH as the dependent variable, we hypothesize a negative sign for the MONTH*RESIGN interaction variable indicating a declining quoted depth for resignation firms relative to dismissal firms.

Change in liquidity. In this analysis, we focus on the change in liquidity in the 2 months following the 8K auditor change filing as compared to the 2 months preceding the filing:[3]

$$\Delta \ln(y_{it}) = \alpha_0 + \alpha_1 \Delta \ln(NTRADE_i) + \alpha_2 \Delta \ln(PRICE_i) + \alpha_3 \Delta \ln(VAR_i) + \alpha_4 UP_i + \alpha_5 DOWN_i + \alpha_6 ACCTG_i + \alpha_7 RESIGN_i + \varepsilon_i \quad (2)$$

In model 2, the subscript i denotes firm. The dependent variable $\Delta \ln(y_i)$ consists of the same three liquidity measures discussed previously except that they are now in the change form (ΔEHS, ΔASCS, and ΔDEPTH). For example, ΔEHS is the log of the ratio of the average effective half-spread in the 2 months *after*

the 8K filing to the average effective half-spread in the 2 months *before* the 8K filing. The first three independent variables (ΔNTRADE, ΔPRICE, and ΔVAR) are the same market microstructure control variables discussed previously except that they too are in the change form. The predicted signs for these control variables are the same as discussed previously. In the regressions with the spread measures (ΔEHS and ΔASCS), the expected signs for ΔNTRADE, ΔPRICE, and ΔVAR are negative, negative, and positive respectively. Similarly, in the regression with DEPTH as the dependent variable, the predicted signs for ΔNTRADE, ΔPRICE, and ΔVAR are positive, negative, and negative, respectively.

In our analysis, we also attempt to control for disclosures in the 8K as to whether the auditor change involved (1) an auditor upgrade or downgrade, or (2) an accounting disagreement between the client and the prior auditor. Variables UP and DOWN are dummy variables indicating whether (based on the 8K disclosures) the auditor change was an upgrade (from a non-Big Six to a Big Six auditor) or a downgrade (from a Big Six to a non-Big Six auditor), respectively. The omitted category consists of auditor changes within the Big Six/non-Big Six, i.e. from a Big Six to another Big Six firm, or from a non-Big Six to another non-Big Six firm. Variable ACCTG is a dummy variable equal to 1 if the 8K reported an accounting disagreement, and 0 otherwise.

To our knowledge, there has been no prior research on the association between market liquidity and type of auditor. Still, since an upgrade (downgrade) to a Big Six (non-Big Six) auditor could potentially imply more (less) credible financial reporting and therefore less information asymmetry, we predict the UP (DOWN) variable to be associated with increased (reduced) liquidity. Keeping in mind that an *increased* spread implies *lower* liquidity, we predict a negative (positive) sign for the UP (DOWN) variable in the regressions with the spread measures (EHS and ASCS) as the dependent variable. Similarly, keeping in mind that a *higher* quoted depth is consistent with *higher* liquidity, we predict a positive (negative) sign for the UP (DOWN) variable in the regression with DEPTH as the dependent variable. Also, since a reported accounting disagreement could signal "opinion shopping" with potential negative implications for information reliability, we predict a negative association between variable ACCTG and liquidity. In the regressions with the spread measures (EHS and ASCS) as the dependent variable, this implies a positive sign for ACCTG. However, in the regression with DEPTH as the dependent variable, we expect a negative sign for ACCTG.

Finally, variable RESIGN is our test variable. This variable is a dummy variable equal to 1 if the 8K disclosed a resignation, and 0 otherwise. As noted previously, the ability of sophisticated investors to anticipate auditor resignations

(based on their ability to extract private information from publicly known litigation risk factors) is expected to generate information asymmetry between these informed investors and others in the months preceding the 8K filing. The public announcement of the auditor resignation in the 8K can therefore be expected to mitigate information asymmetry, which would imply for RESIGN a negative (positive) sign in the regressions with the spread (depth) measures as the dependent variable. However, at this time SEC regulations do not require the client to disclose in the 8K the reasons underlying the auditor resignation. Moreover, as pointed out by Wells and Loudder (1997), professional standards prohibit the auditor from disclosing the private information that motivated the resignation. Given the limited information disclosed in the 8K, the resignation announcement by itself may not be sufficient to reduce information asymmetry. For this reason, we do not predict the sign of variable RESIGN in model 2.

Differential Effects on Trading Activity

In this section, we examine the trading activity associated with the 8K filing to identify the type of investor responding to the auditor resignation/dismissal announcement. Our analysis of trading activity is based on the methodology described in Cready and Mynatt (1991). Basically, we conduct abnormal trading activity tests in which we measure the unexpected number of trades occurring during days −5 to +5 (relative to the 8K filing date) within each of three size-stratified transaction classes, and then test for a cross-sectional association between the unexpected number of trades (within each transaction-size class) and the auditor resignation/dismissal announcement.

Following Cready and Mynatt (1991), the size of the transaction is expected to provide insight into the type of investor responding to the 8K filing. The three transaction classes, adapted from Cready and Mynatt (1991), consist of transactions of: (1) fewer than 300 shares, (2) between 300 and 899 shares, and (3) 2000 shares or more. For brevity, we refer to these size-stratified transaction classes as strata 1, 2, and 3, respectively. Following Cready and Mynatt (1991), transactions in strata 1 and 2 are expected to be initiated by individual investors while transactions in strata 3 (trades of 2000+ shares) are expected to be initiated by institutional investors. As discussed previously, individual investors are not expected to be privy to pre-disclosure private information and are more likely to be dependent on SEC mandated disclosures. Hence, smaller size trades (i.e. transactions classified as belonging to strata 1 and 2) are expected to predominate in the positive abnormal trading activity associated with the 8K auditor resignation filing.

We use the technique reported in Cready and Mynatt (1991) to measure the number of unexpected trades within each transaction-size strata. This technique entails estimating a market model over days −250 to −10 (relative to the 8K auditor change filing date) for each of the three transaction-size strata for each firm.

$$\ln(NTRADE_{ijt}) = \alpha_{ij} + \beta_{ij}\ln(MNTRADE_t) + \sum_{d=-5}^{d=+5} \delta_{dij}D_{dit} + \varepsilon_{ijt}$$
$$\varepsilon_{ijt} = \rho_{ij}\varepsilon_{ijt-1} + \xi_{ijt} \qquad (3)$$

In model 3, subscripts i, j, and t denote firm, trade-size strata, and day, respectively. The log of the number of trades executed daily within each trade-size strata (NTRADE) is regressed on: (1) the log of the mean number of trades executed in all NYSE stocks on that day (MNTRADE), and (2) a vector of dummy variables (D) corresponding to the 11 days in the event period −5 to +5 days surrounding the 8K auditor change filing, where each dummy is equal to 1 if the observation occurred on the event day, and 0 otherwise. Given our three transaction-size strata, there are three market models for each firm. The coefficients on the event dummies measure for each firm's transaction-size strata the unexpected number of trades executed on each event day. Consistent with Cready and Mynatt (1991), we scale these coefficients by their standard error to test the null hypothesis that the standardized abnormal trading activity within each transaction-size strata on each of the 11 days (from day −5 to +5, relative to the 8K filing date) was less than or equal to zero.

Finally, we cumulate the abnormal trading activity over days 0 through +5 (where day 0 is the 8K filing date) to examine the association between cumulative abnormal trading activity (within each transaction-size strata) and the auditor resignation/dismissal announcement.

$$CATA_i = a_0 + a_1UP_i + a_2DOWN_i + a_3ACCTG_i + a_4RESIGN_i + \varepsilon_i \qquad (4)$$

In model 4, subscript i denotes firm. The dependent variable (CATA) is the standardized cumulative abnormal trading activity for each firm in our sample within each of the three transaction-size strata discussed previously. The control variables (UP, DOWN, and ACCTG) were discussed previously in the context of model 2, and attempt to control for disclosures in the 8K as to whether (1) the auditor change involved a switch from a non-Big Six to a Big Six auditor (UP) or from a Big Six to a non-Big Six auditor (DOWN), and (2) there was an accounting disagreement between the client and the prior auditor (ACCTG). Prior research has not examined the association between abnormal trading

activity and these 8K disclosures. Hence, we do not make any predictions regarding the sign of these control variables.

Variable RESIGN is our test variable, which is equal to 1 if the 8K auditor change filing disclosed a resignation, and 0 otherwise. As discussed previously, institutional investors are expected to rely on private information search and to be able to anticipate the auditor resignation announcement. By contrast, individual investors are expected to rely much more heavily on the 8K. As noted previously and consistent with Cready and Mynatt (1991), transactions in strata 1 and 2 are expected to be initiated by individual investors relative to their institutional counterparts. Hence, variable RESIGN is expected to be significant with a positive sign in the regressions for the lower transaction-size strata 1 and 2.

EMPIRICAL FINDINGS

Table 1 provides descriptive data for the variables employed in the analysis and a test for the equality of means between the resignation and dismissal firms. In Table 1, the mean EHS value of 0.027 for the resignation firms indicates that on average the effective half-spread for these firms was equal to 2.7% of share price. The comparison number for the dismissal firms is 1.8% of share price. Since on average the spread measures (EHS and ASCS) are higher and the depth measure (DEPTH) is lower for the resignation firms than for the dismissal firms, the univariate analysis suggests that market liquidity is lower for the resignation firms during the 12 months preceding the 8K filing. A more complete multivariate analysis of this issue is provided in subsequent sections of the paper.[4]

For brevity, we do not present correlation matrices corresponding to the various regressions discussed below. Various diagnostic measures indicated that collinearity is not a significant problem in interpreting our regression results. For example, the highest variance inflation factor (VIF) for any variable in any of our regressions was only 1.25, which is much lower than the threshold level of 10 that might indicate a collinearity problem (Neter et al., 1996). Also, for each of our regressions, we evaluated the sensitivity of our findings to influential observations by calculating the studentized residual for each observation (Belsley et al., 1980). When observations with studentized residuals in excess of the absolute value of 3 were eliminated and the regressions re-estimated, the results were similar to those reported below.[5]

Trend in Liquidity

Table 2 presents the regression estimates using ordinary least squares for model 1. In all three regressions, the control variables NTRADE, PRICE, and VAR

Table 1. Descriptive Statistics.[1]

	Means/(Std. Dev)		
	Resignation (n=22)	Dismissal (n=118)	Test for Equality of Means
EHS	0.027 (0.028)	0.018 (0.019)	t = 12.080***
ASCS	0.016 (0.012)	0.011 (0.0165)	t = 9.230***
DEPTH	27.32 (36.07)	62.93 (203.74)	t = −12.736***
NTRADE	60.26 (183.91)	45.91 (136.41)	t = 5.354***
PRICE	10.33 (14.51)	13.55 (13.93)	t = −7.703***
VAR	0.0009 (0.0016)	0.0006 (0.0018)	t = 4.259***
UP	0.000 (0.000)	0.178 (0.384)	t = −4.357***
DOWN	0.500 (0.512)	0.102 (0.304)	t = 10.042***
ACCTG	0.136 (0.351)	0.017 (0.130)	t = 5.690***

[1] Descriptive statistics are averages for the 12 months preceding the auditor change. Raw measures are reported for descriptive purposes only; the natural log form is used in the regressions (except for variables UP, DOWN, and ACCTG).

EHS: monthly average effective half-spread. The effective half-spread is defined as $|p_t - m_t|$ where p_t is the transaction price at time t and m_t is the midpoint (average) of bid and ask quotes at time t, deflated by the midpoint of the quoted bid and ask quotes.

ASCS: monthly average adverse selection component of the spread. The adverse selection component is the excess of the effective half-spread over the realized half-spread deflated by the midpoint of the quoted bid and ask prices.

DEPTH: monthly average quoted depth, i.e. the number of hundred share lots that the market maker is willing to trade at the quoted bid and ask prices, calculated as the sum of the depth at the ask price and the depth at the bid price.

NTRADE: monthly average of the number of trades executed per day.

PRICE: monthly average of the bid and ask quotes.

VAR: monthly average of the daily variance in intraday stock returns.

UP: dummy variable equal to 1 if the firm changed from a non-Big 6 to a Big 6 auditor, 0 otherwise.

DOWN: dummy variable equal to 1 if the firm changed from a Big 6 to a non-Big 6 auditor, 0 otherwise.ACCTG: dummy variable equal to 1 if the 8K disclosed an accounting disagreement between the client and auditor, 0 otherwise.

Table 2. Test for Trend in Liquidity in the 12 Months Preceding the 8K Auditor Change Filing.

$$\ln(y_{it}) = \alpha_{i0} + \alpha_1 \ln(NTRADE_{it}) + \alpha_2 \ln(PRICE_{it}) + \alpha_3 \ln(VAR_{it}) + \alpha_4 MONTH_{it} + \alpha_5(MONTH_{it} * RESIGN_{it}) + \varepsilon_{it} \quad (1)$$

Dependent variable	α_1	α_2	α_3	α_4	α_5	Adj. R^2
EHS	−0.144***	−0.535***	0.098***	−0.007***	0.010***	0.419
	(−9.195)	(−21.415)	(10.799)	(−3.802)	(2.281)	
ASCS	−0.210***	−0.564***	0.020	−0.010***	0.029***	0.159
	(−6.646)	(−11.193)	(1.112)	(−2.958)	(3.439)	
DEPTH	0.094***	−0.192***	−0.025***	0.002	−0.017***	0.039
	(5.307)	(−6.780)	(−2.459)	(0.978)	(−3.525)	

EHS: monthly average effective half-spread. The effective half-spread is defined as $|p_t - m_t|$ (where p_t is the transaction price at clock time t and m_t is the midpoint of the bid and ask quotes at clock time t) deflated by the midpoint of the bid and ask quotes.
ASCS: monthly average adverse selection component of the spread. The adverse selection component is the excess of the effective half-spread over the realized half-spread deflated by the midpoint of the bid and ask quotes.
DEPTH: monthly average quoted depth, i.e., the number of hundred share lots that the market maker is willing to trade at the quoted bid and ask prices, calculated as the sum of the depth at the ask price and the depth at the bid price.
NTRADE: monthly average of the number of trades executed per day.
PRICE: monthly average of the bid and ask quotes.
VAR: monthly average of the daily variance in intraday stock returns.
MONTH: equal to 0 in month −12 relative to the 8K auditor change filing month, 1 for month −11, 2 for month −10, and so on up to the month preceding the 8K filing.
MONTH*RESIGN: interaction variable between MONTH and RESIGN, where RESIGN is a dummy variable equal to 1 for an auditor resignation firm, 0 otherwise.
The model is a fixed effects panel model estimated across a panel data set consisting of 12 monthly observations preceding the month of the 8K auditor change filing for each of the 140 auditor change firms. Significance levels of 0.10, 0.05, and 0.01 are denoted by *, **, and ***, respectively, and are based on one-tailed tests where the sign of the coefficient is predicted and two-tailed otherwise. The subscripts i and t denote firm and month, respectively (except in the definition of the effective half-spread, $|p_t - m_t|$, where the subscript t denotes clock time). For brevity, we do not report estimates of the firm-specific intercept α_{i0}.

have the expected signs consistent with the prior market microstructure literature and imply plausible relationships between the three dependent variables and the control variables. In particular, our results show that: (1) the number of trades (NTRADE) is associated negatively with spreads and positively with quoted depth, (2) the price of the stock (PRICE) is associated negatively with spreads and quoted depth, and (3) the return variance (VAR) is associated positively with spreads and negatively with depth.

In the regressions with the spread measures (EHS and ASCS) as the dependent variables, the variable MONTH is significant with a *negative* sign indicating that spreads were declining (i.e. liquidity was increasing) over the 12 month period for our sample firms.[6] However, the interaction variable MONTH*RESIGN (which is our test variable) is significant with the expected *positive* sign; moreover, the coefficient value of MONTH*RESIGN exceeds that of the MONTH variable. Summing the coefficients for the MONTH and the MONTH*RESIGN variables, the net coefficient value is positive which indicates that for the resignation firms spreads were increasing (i.e. liquidity was decreasing) over the 12 months prior to the 8K filing, although for the dismissal firms spreads were declining over that same period consistent with the trend in the overall market (see note 6).

In the regression with DEPTH as the dependent variable, variable MONTH is not significant. However, the test interaction variable MONTH*RESIGN is significant with the expected *negative* sign indicating that for the resignation firms the quoted depth was decreasing during the 12 months preceding the 8K filing. Holding other things constant, an increase in spread or a decrease in quoted depth are both consistent with lower liquidity. Hence, our results for all three regressions are consistent in indicating that for the resignation firms (but *not* for the dismissal firms) liquidity was decreasing during the 12 months leading up to the 8K filing.[7]

Change in Liquidity

Table 3 presents the regression estimates using ordinary least squares for model 2. Once again, our regression results yield estimates consistent with the prior literature and with expected signs for the market microstructure control variables (NTRADE, PRICE, and VAR). In particular, the results indicate that: (1) a change in the number of trades (NTRADE) is associated negatively (positively) with a change in spreads (quoted depth), (2) a change in the price of the stock (PRICE) is associated negatively with a change in both spreads and quoted depth, and (3) a change in the return variance (VAR) is associated positively with a change in the effective spread but not quoted depth.

Table 3. Test for Trend in Liquidity Around the 8K Auditor Change Filing.

$$\Delta \ln(y_i) = \alpha_0 + \alpha_1 \Delta \ln(NTRADE_i) + \alpha_2 \Delta \ln(PRICE_i) + \alpha_3 \Delta \ln(VAR_i) + \alpha_4 UP_i + \alpha_5 DOWN_i\ \alpha_6 ACCTG_i + \alpha_7 (RESIGN_i + \varepsilon_i \quad (2)$$

Dependent variable	α_0	α_1	α_2	α_3	α_4	α_5	α_6	α_7	Adj. R^2
ΔEHS	−0.005	−0.124**	−0.477***	−0.076***	0.025	0.077	−0.088	0.061	0.277
	(−0.188)	(−2.075)	(−5.174)	(2.196)	(0.465)	(1.298)	(−0.899)	(1.031)	
ΔASCS	−0.010	−0.074	−0.462***	−0.038	0.023	0.323***	−0.289	−0.042	0.095
	(−0.207)	(−0.647)	(−2.613)	(−0.581)	(0.225)	(2.828)	(−1.544)	(−0.377)	
ΔDEPTH	−0.017	−0.088*	0.015*	−0.032	−0.005	0.051	−0.036	0.000	0.000
	(−0.679)	(1.413)	(−1.613)	(−0.888)	(−0.089)	(0.816)	(−0.349)	(0.002)	

Δ ln(EHS): change in the natural log of the average effective half-spread in the 2 months following the auditor change as compared to the 2 months preceding the auditor change. The effective half spread is defined as $|p_t - m_t|$ (where p_t is the transaction price at clock time t and m_t is the midpoint of the bid and ask quotes at clock time t) deflated by the midpoint of the bid and ask quotes.

Δ ln(ASCS): change in the natural log of the average adverse selection component of the spread in the 2 months following the auditor change as compared to the 2 months preceding the auditor change. The adverse selection component of the spread is measured as the of the excess of the effective half-spread over the realized half-spread deflated by the midpoint of the bid and ask quotes.

Δ ln(DEPTH): change in the natural log of the quoted depth in the 2 months following the auditor change as compared to the 2 months preceding the auditor change. Quoted depth is the average number of hundred share lots that the market maker is willing to trade at the quoted bid and ask prices calculated as the sum of depth at the ask price and the depth at the bid price.

Δ ln(NTRADE): change in the natural log of the average daily number of trades in the 2 months following the auditor change as compared to the 2 months preceding the auditor change.

Δ ln(PRICE): change in the natural log of the average of the bid and ask quotes in the 2 months following the auditor change as compared to the 2 months preceding the auditor change.

Δ ln(VAR): change in the natural log of average daily variance in intraday stock returns in the 2 months following the auditor change as compared to the 2 months preceding the auditor change.

UP: dummy variable equal to 1 if the firm changed from a non-Big 6 to a Big 6 auditor, 0 otherwise.

DOWN: dummy variable equal to 1 if the firm changed from a Big 6 to a non-Big 6 auditor, 0 otherwise.

ACCTG: dummy variable equal to 1 if the 8K disclosed an accounting disagreement between the client and auditor, 0 otherwise.

RESIGN: dummy variable equal to 1 if the firm is a member of the resignation group, 0 otherwise.

The model is estimated cross-sectionally across the sample of 140 auditor change firms. Significance levels of 0.10, 0.05, and 0.01, denoted by *, **, and ***, respectively, are based on one-tailed tests where the sign of the coefficient is predicted and two -tailed otherwise. The subscript i denotes firm.

In model 2, we also attempted to control for disclosures in the 8K as to the direction of the auditor change and reported accounting disagreements. The results in Table 3 suggest that an auditor upgrade from a non-Big Six to a Big Six auditor (variable UP) is not associated with a change in either spread or depth. Similarly, a reported accounting disagreement between the client and the prior auditor (variable ACCTG) is not associated with a change in either measure of market liquidity. However, an auditor downgrade from a Big Six to a non-Big Six auditor (variable DOWN) is associated with a change in the adverse selection component of the spread (ASCS) with the expected positive sign but not with a change in the other liquidity measures (EHS and DEPTH).

Finally, the test variable (RESIGN) is not significant in any of the regressions. For the resignation firms, although liquidity was decreasing in the 12 months leading up to the announcement, there appears to be no change in liquidity in the 2 months subsequent to the 8K filing as compared to the 2 months preceding the filing.[8] This suggests that the auditor resignation announcement halts the trend towards lower liquidity but does not improve liquidity. As noted previously, the client is not required to disclose the reasons motivating the resignation and professional standards prohibit the auditor from disclosing private information obtained during the audit.

Hence, the implications of the resignation announcement for investors are subject to interpretation (an area of activity where once again informed investors would have an advantage) and could explain the persistence of information asymmetry and lower liquidity even after the 8K filing.

Abnormal Trading Activity

Table 4 presents the mean standardized abnormal trading activity for our sample firms over the 11 days (−5, +5) surrounding the 8K auditor resignation/dismissal filing for each of the three transaction-size classes.[9] For dismissal announcements, little positive abnormal trading activity is to be observed. The strongest response for dismissal announcements occurs on day +3 for transaction-size strata 3. However, cumulatively for days (0, +1) and (0, +5) there is no evidence of positive abnormal trading activity for dismissal firms. As discussed previously, prior market-based research indicates no significant price reaction on average to auditor dismissals, which suggests that investors perceive these dismissals to be legitimately motivated rather than to represent opportunistic (opinion shopping) behavior on the part of management. In Table 4, the near absence of abnormal trading activity subsequent to the 8K auditor dismissal filing for all transaction-size strata is consistent with the evidence from prior research using alternative methodologies.

Table 4. Mean Standardized Daily Abnormal Trading Activity By Trade-Size Strata for Resignation/Dismissal Firms (z-statistic).

$$\ln(NTRADE_{ijt}) = \alpha_{ij} + \beta_{ij}\ln(MNTRADE_t) + \sum_{d=-5}^{d=+5} \delta_{di}D_{di} + \varepsilon_{ijt},$$
$$\text{where } \varepsilon_{ijt} = \rho_{ij}\varepsilon_{ijt-1} + \xi_{ijt} \qquad (3)$$

	Trade-Size Strata					
	Strata 1 (<300 shares)		Strata 2 (300–899 shares)		Strata 3 (2000+ shares)	
	Resign.	Dismissal	Resign.	Dismissal	Resign.	Dismissal
$\bar{\delta}_{-5j}$	−0.407	−0.087	−0.365	0.050	−0.426	−0.039
$\bar{\delta}_{-4j}$	0.055	−0.134	−0.036	0.036	0.011	0.159*
$\bar{\delta}_{-3j}$	0.664***	0.05	0.419*	0.030	0.255	−0.141
$\bar{\delta}_{-2j}$	0.185	0.105	0.115	−0.196	−0.057	0.030
$\bar{\delta}_{-1j}$	0.297	−0.025	0.188	−0.045	0.104	−0.010
$\bar{\delta}_{0j}$	0.503**	−0.171	0.166	0.133	0.049	−0.015
$\bar{\delta}_{1j}$	0.176	−0.127	0.759***	−0.140	0.218	−0.195
$\bar{\delta}_{2j}$	0.927***	0.129	0.745***	−0.181	0.241	0.027
$\bar{\delta}_{3j}$	0.848***	−0.055	0.863***	0.037	0.560**	0.213**
$\bar{\delta}_{4j}$	0.636**	0.163*	0.816***	0.074	−0.046	0.064
$\bar{\delta}_{5j}$	0.106	−0.171	0.731**	−0.260	−0.120	0.106
(0, +1)	679*	−0.298	0.925**	−0.007	0.267	−0.210
(0, +5)	3.196***	−0.232	4.08***	−0.337	0.902	0.200

where $NTRADE_{ijt}$ is the number of trades within strata j in firm i on day t; $MNTRADE_t$ is the average number of trades in NYSE stocks on day t; D_{di} is a dummy variable assigned a value of 1 if the observation in firm i occurs on the dth day during the event window; $\bar{\delta}_{dj}$ denotes the mean level of standardized abnormal trading on event day d within strata j. Reported significance levels of *, **, and *** (denoting 0.10, 0.05 and 0.01 significance levels) are from a test of the null hypothesis that the cross-sectional mean of the standardized abnormal trading activity is less than or equal to zero.

In contrast to auditor dismissals, auditor resignations appear to generate a higher than expected number of trades (positive abnormal trading activity) on a number of different event days and across all transaction-size strata. In particular, the positive abnormal trading activity appears to be concentrated on days +1 through +5. Klock (1994) has indicated that the 8K is unavailable in the SEC's Public Reference Room for up to 3 days after the filing date. Similarly, EDGAR filings are unavailable until the day following submission. This delay in disseminating the 8K might explain the near absence of abnormal trading activity on day 0.

In examining the event days (0,+5) subsequent to the 8K resignation filing, the positive abnormal trading activity appears to be concentrated among the smaller transaction-size strata 1 and 2. Following Cready and Mynatt (1991), the size of the transaction is expected to provide insight into the type of investor responding to the 8K filing. Thus, trades in strata 1 and 2 are expected to be initiated by individual investors. In contrast, trades in transaction-size strata 3 (i.e. trades of 2000+ shares) are expected to be initiated by institutional investors.

For strata 3, positive abnormal trading activity associated with an auditor resignation is to be observed on day +3 and is significant at the 0.05 level. However, the bulk of the positive abnormal trading activity associated with resignations is to be observed for strata 1 and 2, and is generally significant at the 0.01 level. The results suggest that smaller transactions predominate in the positive abnormal trading activity associated with the 8K auditor resignation filing. As discussed previously, institutional investors are expected to be able to afford private information search and to be able to anticipate auditor resignations. By contrast, individual investors are less likely to be privy to private pre-disclosure information and are expected to be much more dependent on SEC mandated disclosure. Thus, in Table 4 positive abnormal trading activity subsequent to the 8K filing is concentrated in the lower transaction-size strata 1 and 2 and only for auditor resignations.

Table 5 provides additional evidence on abnormal trading activity subsequent to the 8K resignation filing and transaction-size strata. The regression estimates in Table 5 are for model 4 and for each of the three transaction-size classes. The dependent variable (CATA) represents the cumulative abnormal trading activity on days 0 through +5 relative to the 8K auditor change filing. Prior research (e.g. Cready & Mynatt, 1991) has not identified control variables for cumulative abnormal trading activity. In Table 5, we control for disclosures in the 8K regarding the direction of the auditor change (such as an upgrade to a Big Six from a non-Big Six auditor, or a downgrade to a non-Big Six from a Big Six auditor) and the presence (or lack thereof) of an accounting disagreement between the client and the predecessor auditor. For transaction-size strata

Table 5. Test for an Association Between Cumulative Abnormal Trading Activity (By Trade-Size Strata) and Resignation/Dismissal Group.

$$CATA_i = \alpha_0 + \alpha_1 UP_i + \alpha_2 DOWN_i + \alpha_3 ACCTG_i + \alpha_4 RESIGN_i + \varepsilon_i$$

	Trade-Size Strata		
	Strata 1 (< 300 shares)	Strata 2 (300–899 shares)	Strata 3 (> 1999 shares)
α_0	0.104	-0.081	0.131
	(0.598)	(-0.506)	(0.816)
α_1	-0.685*	0.155	-0.361
	(-1.676)	(0.437)	(-1.015)
α_2	-0.574	-0.634*	-0.273
	(-1.342)	(-1.694)	(-0.716)
α_3	-0.575	-0.226	0.192
	(-0.717)	(-0.306)	(0.290)
α_4	1.269***	1.564***	0.238
	(2.630)	(3.906)	(0.609)

CATA is the standardized cumulative abnormal trading activity occurring during the event window (days 0 through +5 relative to the 8K auditor change filing date), calculated as:

$$\frac{1}{\sqrt{6}} \sum_{d=0}^{5} \frac{\delta_{dij}}{\sqrt{Var(\delta_{dij})}}$$

and δ_{dij} is the measure of abnormal trading activity in strata j for firm i on event day d as obtained from model 3.

UP: dummy variable equal to 1 if the firm changed from a non-Big 6 to a Big 6 auditor, 0 otherwise.
DOWN: dummy variable equal to 1 if the firm changed from a Big 6 to a non-Big 6 auditor, 0 otherwise.
ACCTG: dummy variable equal to 1 if the 8K disclosed an accounting disagreement between the client and auditor, 0 otherwise.
RESIGN: dummy variable equal to 1 if the firm is a member of the resignation group, 0 otherwise.

Significance levels of 0.10, 0.05, and 0.01, denoted by *, **, and ***, respectively, are based on one-tailed tests where the sign of the coefficient is predicted and two-tailed otherwise. The subscript i denotes firm.

1 and 2, the test variable RESIGN is significant at the 0.01 level with the expected positive sign indicating that the 8K auditor resignation filing triggered a higher than expected number of trades (positive abnormal trading activity) among individual investors but not among institutional investors.[10]

CONCLUDING REMARKS

The prior literature (e.g. Lev, 1988; Levitt, 1998; Shefrin & Statman, 1992) suggests that SEC disclosure regulations are motivated primarily by its mandate to protect investors and to preserve public confidence in the capital markets as a level playing field by reducing information asymmetry between informed investors and others. Information asymmetry can have deleterious effects on market liquidity as measured by bid-ask spreads and quoted depths. Although the SEC has long required that auditor changes be made public in the Form 8K, mandatory disclosure of whether the auditor resigned or was dismissed is relatively recent. In this paper, we examined the differential effects on market liquidity and trading activity (across different investor groups) associated with auditor resignations and dismissals. Since our analysis was based on a limited sample of 22 resignation observations and 118 dismissal observations, the results must be interpreted with caution.

In our analysis, we could not reject the null hypothesis that auditor dismissals have no effect in terms of either market liquidity or trading activity. These findings are consistent with prior research (using alternative methodologies) which indicate that the 8K auditor dismissal filing is *not* a significant information event. By contrast, for auditor resignations, our results suggest that the 12 month period preceding the 8K resignation filing is characterized by rising information asymmetry. Also, the results suggest that the resignation announcement is a significant information event for individual investors (as indicated by the abnormal increase in trading activity subsequent to the 8K filing for lower size transactions). However, we were unable to reject the null that the resignation announcement is not an information event for institutional investors. These findings are consistent with the notion that informed investors may be able to anticipate auditor resignations based on publicly known litigation risk factors as discussed by Krishnan and Krishnan (1997).[11]

From a policy perspective, the overall results suggest that the SEC's decision to differentiate between auditor resignations and dismissals in the 8K is informative for individual investors. However, in our analysis, we could not reject the null that the resignation announcement has no effect on information asymmetry subsequent to the 8K filing. Institutional investors may have some knowledge of the circumstances motivating the resignation (based on their ability to acquire private pre-disclosure information). By contrast, individual investors are informed of the auditor resignation by the 8K but not of the reasons underlying the resignation. Current SEC disclosure standards do not require the client to disclose the reasons for the resignation and professional standards prohibit the auditor from disclosing the private information (acquired during

the audit) that may have triggered the resignation. Possibly for these reasons, the 8K filing appears to halt the trend towards increasing information asymmetry during the 12 months preceding the resignation announcement, but does not actually reduce information asymmetry subsequent to the filing.

NOTES

1. The SEC has sought to impede "opinion shopping" motivated auditor changes by requiring communication between the successor and the prior auditor, and timely disclosure (within five business days) of: (1) the auditor change, and (2) technical disagreements (if any) between the client and the prior auditor in the preceding two years. See Krishnan and Stephens (1995) and Smith (1986) for a discussion of the opinion shopping literature.

2. By contrast, DeFond et al. (1997) report a significant negative price reaction for the resignation firms (relative to a control sample of dismissal firms) in the pre-filing period, i.e. the period between the auditor change and the day prior to the 8K filing. They (p. 39) suggest that this may be due to "insider leakage" of the upcoming filing.

3. As discussed in footnote 8 below, we also examine the change in liquidity using a narrower event window surrounding the 8K filing.

4. In our sample of 140 auditor change firms, the number of auditor upgrades (non-Big Six to Big Six), downgrades (Big Six to non-Big Six), and reported accounting disagreements between the client and the prior auditor were 21, 23, and 4, respectively. As noted previously, DeFond and Jiambalvo (1993) suggest that accounting disagreements may be under-reported since such disclosure may not be in the best interest of either the client or the auditor.

5. As discussed previously, the focus of our study is information asymmetry. However, for completeness (as suggested by an anonymous reviewer), we also tested the price response to the 8K filing for our auditor resignation and dismissal firms. These tests indicated a statistically significant and negative abnormal return on day +1 for the resignation firms but not the dismissal firms. Also, the cumulative abnormal return over days (0, +5) was statistically significant and negative for the resignation (but not dismissal) firms. These results are consistent with prior research by DeFond et al. (1997) and Wells and Louder (1997).

6. The trend towards lower spreads for our sample firms may simply be reflective of declining spreads in the overall market during the period examined. To test this hypothesis, we re-estimated model 1 for the population of all firms ($n = 6{,}131$) in the TAQ database for the period 1994–1996. In the regressions with the spread measures (EHS and ASCS) as the dependent variable, the slope coefficient on MONTH was *negative* and significant at the 0.01 level. Hence, the trend towards lower spreads in our sample appears to reflect the overall market trend during this time period.

7. As suggested by an anonymous reviewer, we evaluated the sensitivity of the reported results to the effects of size and financial distress by re-estimating model 1 separately for data partitions formed on the basis of whether a firm was above or below the sample median value of size and whether a firm was above or below the sample median measure of financial distress. Size was measured by the market value of equity (calculated as of the prior fiscal year-end), and financial distress was measured by the Z-score obtained

from Altman's (1968) bankruptcy prediction model based on data from the annual report issued immediately prior to the auditor change. Results obtained from these partitioned analyses were entirely consistent with the results reported in the paper. Thus, the sensitivity analysis indicated that the reported results are *not* sensitive to the effects of size or financial distress.

8. As additional analysis, we re-estimated model 2 using a much narrower 11 day event window by examining the change in market liquidity in the 6 days (0, +5) after the 8K filing as compared to the 5 days (−5, −1) before the filing. The results for the re-estimated model were similar to those reported for the longer (2 months after as compared to 2 months before) event window. The longer 4 month event window reported in the text of the paper allows for any delayed effects on market liquidity resulting from the 8K filing.

9. As suggested by an anonymous reviewer, we investigated whether auditor resignations were associated with delays in filing the Form 8K. For each of the 140 firms in our dataset, we calculated the number of business days after the auditor change that the 8K was filed. Any 8K that was filed more than five business days after the auditor resignation/dismissal was characterized as a late filer. Six resignation firms and twenty-six dismissal firms were late filers. A chi-square test found no association between being a late filer and a resignation (chi-square statistic = 0.289, p-value = 0.591). We also tested for an association between the length of the filing delay and the abnormal trading activity observed within each of the size stratified transaction classes. We found no such association. As pointed out by an anonymous reviewer, Carter and Soo (1999) find some evidence of a stock price reaction to timely 8K filings related to auditor changes.

10. As suggested by an anonymous reviewer, we also conducted mean transaction size analysis. The results were weaker than, but consistent with, the results reported in Tables 4 and 5. In particular, transactions executed during days (0, +5) were significantly *smaller* than expected for the resignation (but not dismissal) firms (p-value = 0.06), indicating increased trading activity among individual investors relative to the trading activity of institutional investors for resignation (but not dismissal) firms. We believe that the size-stratified analysis reported in the paper represents a methodo-logical advance over the mean transaction size analysis used in Cready (1988). A potential disadvantage of mean transaction size analysis is that it can be distorted by a few large transactions. Also, as discussed by Cready and Mynatt (1991, p. 306), number of transactions is a more sensitive indicator of abnormal trading activity than the size of the trade (number of shares traded). For these reasons, we have reported the results of the size-stratified analysis in the paper.

11. As noted previously, because a reported resignation could cause a stock price decline and trigger a lawsuit against both the client and the auditor, Krishnan and Krishnan (1997, fn. 15) suggest that both parties have an incentive to avoid reporting a resignation. For this reason, resignations are likely to be under-reported with some resignations misclassified as dismissals. Such misclassifications would in fact bias our study *against* finding the differential effects that we report. However, as pointed out by an anonymous reviewer and supported by Shu (1999), a resignation is not necessarily indicative of future investor losses since some resignations can result from an auditor-client mismatch unrelated to litigation risk or financial distress. Changes in auditor specialization and technology may cause a client to be mismatched with an auditor's targeted clientele. Hence, the auditor realignment (whether initiated by the client or the auditor) may not be indicative of a negative development.

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REFERENCES

Altman, R. (1968). Financial ratios, discriminant analysis and the prediction of corporate bankruptcy. *Journal of Finance,* (September), 589–609.

Amihud, Y., & Mendelson, H. (1986). Asset pricing and the bid-ask spread. *Journal of Financial Economics, 17,* 223–249.

Bagehot, W. (1971). The only game in town. Financial Analysts Journal (March/April): 12-14.

Beaver, W. (1998). Financial Reporting: An Accounting Revolution. Third Edition. Upper Saddle River, NJ: Prentice Hall.

Bedingfield, J., & Loeb, S. (1974). Auditor changes: an examination. *Journal of Accountancy,* (March), 66–69.

Belsley, D., Kuh, E., & Welsch, R. (1980). *Regression Diagnostics, Identifying Influential Data and Sources of Collinearity.* New York: John Wiley.

Bockus, K., & Gigler, F. (1998). A theory of auditor resignation. *Journal of Accounting Research,* (Autumn), 191–208.

Brennan, M., & Subrahmanyam, A. (1995). Investment analysis and price formation in securities markets. *Journal of Financial Economics, 38,* 361–381.

Burton, J., & Roberts, W. (1967). A study of auditor changes. *Journal of Accountancy,* (April), 31–36.

Business Week. (1993). Big Six firms are firing clients. (March 1), 76–77.

Callahan, C., Lee, C., & Yohn, T. (1997). Accounting information and bid-ask spreads. *Accounting Horizons,* (December), 50–60.

Carter, M., & Soo, B. (1999). The relevance of form 8K reports. *Journal of Accounting Research,* (Spring), 119–132.

Chow, C., & Rice, S. (1982). Qualified audit opinions and auditor switching. *The Accounting Review,* (April), 326–335.

Coller, M., & Yohn, T. (1997). Management forecasts and information asymmetry: an examination of bid-ask spreads. *Journal of Accounting Research,* (Autumn), 181–191.

Copeland, T., & Galai D. (1983). Information effects and the bid-ask spread. *Journal of Finance, 38*(5), 1457–1469.

Cready, W. (1988). Information value and investor wealth: The case of earnings announcements. *Journal of Accounting Research,* (Spring), 1–27.

Cready, W., & Mynatt, P. (1991). The information content of annual reports: a price and trading response analysis. *The Accounting Review,* (April), 291–312.

DeFond, M., & Jiambalvo, J. (1993). Factors related to auditor-client disagreements over income-increasing accounting methods. *Contemporary Accounting Research,* (Spring), 415–431.

DeFond, M., Ettredge, M., & Smith, D. (1997). An investigation of auditor resignations. *Research in Accounting Regulation,* (11), 25–45.

Easley, D., Kiefer, N., O'Hara, M., & Paperman, J. (1996). Liquidity, information, and infrequently traded stocks. *Journal of Finance*, (September), 1405–1436.

Eichenseher, J., & Shields, D. (1983). The correlates of CPA-firm change for publicly-held corporations. *Auditing: A Journal of Practice & Theory*, *2*(2, Spring), 23–37.

Eichenseher, J., Hagigi, M., & Shields, D. (1990). Market reaction to auditor changes by OTC companies. *Auditing: A Journal of Practice & Theory*, (Fall), 29–40.

Fried, D., & Schiff, A. (1981). CPA switches and associated market reactions. *The Accounting Review*, *56*(2, April), 326–341.

Glosten, L., & Harris, L. (1988). Estimating the components of the bid/ask spread. *Journal of Financial Economics*, *21*, 123–142.

Glosten, L., & Milgrom, P. (1985). Bid-ask and transaction prices in a specialist market with heterogeneously informed traders. *Journal of Financial Economics*, (March), 71–100.

Hakansson, N. (1977). Interim disclosure and public forecasts: an economic analysis and a framework for choice. *The Accounting Review*, (April), 396–416.

Hakansson, N. (1981). On the politics of accounting disclosure and measurement: an analysis of economic incentives. *Journal of Accounting Research*, (Supplement), 1–35.

Huang, R., & Stoll, H. (1996). Dealer versus auction markets: A paired comparison of execution costs on NASDAQ and the NYSE. *Journal of Financial Economics*, *41*, 313–357.

Ip, G. (1998). Individuals role in stock market grows as the influence of institutions declines. *Wall Street Journal*, (November 16), C1.

Johnson, W., & Lys, T. (1990). The market for audit services: evidence from voluntary auditor changes. *Journal of Accounting and Economics*, *12*, 281–308.

Kim, O., & Verrecchia, R. (1994). Market liquidity and volume around earnings announcements. *Journal of Accounting Economics*, (January), 41–68.

Klock, M. (1994). The stock market reaction to a change in certifying accountant. *Journal of Accounting, Auditing, and Finance*, (Spring), 339–347.

Knapp, M., & Elikai, F. (1988). Auditor changes: A note on the policy implications of recent analytical and empirical research. *Journal of Accounting, Auditing, and Finance*, (Winter), 78–86.

Krishnan, J., & Krishnan, J. (1997). Litigation risk and auditor resignations. *The Accounting Review*, *72*(4) October, 539–560.

Krishnan, J., & Stephens, R. (1995). Evidence on opinion shopping from audit opinion conservatism. *Journal of Accounting and Public Policy*, (Fall), 179–201.

Lee, C., & Ready, M. (1991). Inferring trade direction from intraday data. *Journal of Finance*, *46*(2) (June), 733–746.

Lee, C., Mucklow, B., & Ready, M. (1993). Spreads, depths, and the impact of earnings information: An intraday analysis. *Review of Financial Studies*, 345–375.

Lev, B. (1988). Toward a theory of equitable and efficient accounting policy. *The Accounting Review*, (January), 1–22.

Levitt, A. (1998). The importance of high quality accounting standards. *Accounting Horizons*, (March), 79–82.

Lys, T., & Watts, R. (1994). Lawsuits against auditors. *Journal of Accounting Research*, *32*, (Supplement), 65–93.

MacDonald, E. (1997). More accounting firms are dumping risky clients; Big Six worry about their reputations and chance of costly litigation. *Wall Street Journal*, (April 25), A2.

McConnell, D. (1984. Auditor changes and related disagreements. *Auditing: A Journal of Practice & Theory*, *3*(2), (Spring), 44–56.

Menon, K., & Williams, D. (1999). Error cost and auditors' termination decisions. *Journal of Accounting Auditing and Finance*, (Spring), 95–123.

Neter, J., Kutner, M., Nachtsheim, C., & Wasserman, W. (1996). *Applied Linear Statistical Models*. Chicago, IL: Irwin.

Nichols, D., & Smith, D. (1983). Auditor credibility and auditor changes. *Journal of Accounting Research*, *21*(2), (Autumn), 534–544.

O'Hara, M. (1995). *Market Microstructure Theory*. Cambridge, MA: Blackwell Publishers.

Phillips, S., & Zecher, J. (1981). *The SEC and the Public Interest*. Cambridge, MA: The MIT Press.

Schwartz, K., & Menon, K. (1985). Auditor switches by failing firms. *The Accounting Review*, (April), 248–261.

Shefrin, H., & Statman, M. (1992). *Ethics, Fairness, Efficiency, and Financial Markets*. Charlottesville, VA: The Research Foundation of the Institute of Chartered Financial Analysts.

Shu, S. (1999). *Auditor Resignations: Clientele Effects and Legal Liability*. Unpublished dissertation. (University of Rochester).

Smith, D. (1986). Auditor "subject to" opinions, disclaimers, and auditor changes. *Auditing: A Journal of Practice and Theory*, (Fall), 95–108.

Stice, J. (1991). Using financial and market information to identify pre-engagement factors associated with lawsuits against auditors. *The Accounting Review*, (July), 516–533.

Stoll, H. (1976). Dealer inventory behavior: an empirical investigation of NASDAQ stocks. *Journal of Financial and Quantitative Analysis*, September, *11*(3), 359–380.

Stoll, H. (1978). The pricing of security dealer services: An empirical study of NASDAQ stocks. *The Journal of Finance*, September, *33*(4), 1153–1172.

Stoll, H. (1989. Inferring the components of the bid-ask spread: theory and empirical tests. *The Journal of Finance*, March, *44*(1), 115-134.

Teoh, S. (1992). Auditor independence, dismissal threats, and the market reaction to auditor switches. *Journal of Accounting Research*, *30*(1), (Spring), 1–23.

Venkatesh, P., & Chiang, R. (1986). Information asymmetry and the dealer' bid-ask spread: a case study of earnings and dividend announcements. *Journal of Finance*, (December), 1089–1102.

Wells, D., & Loudder, M. (1997). The market effects of auditor resignations. *Auditing: A Journal of Practice & Theory*, *16*(1), (Spring), 138–144.

White, H. (1980). A heteroscedasticity-consistent covariance matrix estimator and a direct test for heteroscedasticity. *Econometrica*, *48*(4), 817–838.

Zuckerman, G., & Ip, G. (1998). It isn't just investors who are smarting from liquidity crunch. *Wall Street Journal*, (November 6), A1.